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	Reference	Dataset	Type of dataset	Tool	TP	FP	TP+FP	PP	Sensitivity	Precision	F1	Execution time (per run) [h]	Memory usage (per sample)
 13	[1]	set1.1	simulated	Arriba	90	7	97	150	0,60	0,93		1	38
	[1]	set1.1	simulated	STAR-Fusion	52	13	65	150	0,35	0,80		2	10
	[1]	set1.1	simulated	FusionCatcher				150				6	38
	[1]	set1.1	simulated	defuse				150				8	5
	[1]	set1.1	simulated	InFusion	20	17	37	150	0,13	0,54		14	5
	[1]	set1.1	simulated	SOAPfuse	56	12	68	150	0,37	0,82		17	7
	[1]	set1.1	simulated	PRADA				150				27	7
	[1]	set1.2	spike-in	Arriba	10	0	10	20	0,50	1,00		0,25	38
	[1]	set1.2	spike-in	FusionCatcher	8	32	40	20	0,40	0,20		1	28
	[1]	set1.2	spike-in	STAR-Fusion	7	0	7	20	0,35	1,00		1	38
	[1]	set1.2	spike-in	defuse	7	42	49	20	0,35	0,14		2,5	6
	[1]	set1.2	spike-in	InFusion	5	75	80	20	0,25	0,06		4,5	5
	[1]	set1.2	spike-in	SOAPfuse	8	25	33	20	0,40	0,24		5	6
 15	[1]	set1.2	spike-in	PRADA				20				9	15
	[1]	set1.3	real (cells)	Arriba	78	80	158	80	0,98	0,49		1	39
	[1]	set1.3	real (cells)	STAR-Fusion	38	10	48	80	0,48	0,79		2	40
	[1]	set1.3	real (cells)	FusionCatcher	42	200	242	80	0,53	0,17		20	28
	[1]	set1.3	real (cells)	defuse	70	280	350	80	0,88	0,20		25	5
	[1]	set1.3	real (cells)	InFusion	60	350	410	80	0,75	0,15		48	8
	[1]	set1.3	real (cells)	PRADA				80				50	18
	[1]	set1.3	real (cells)	SOAPfuse	44	280	324	80	0,55	0,14		148	10
	[1]	set1.4	real	Arriba	55			56	0,98				
	[1]	set1.4	real	FusionCatcher	54			56	0,96				
	[1]	set1.4	real	InFusion	49			56	0,88				
	[1]	set1.4	real	PRADA	49			56	0,88				
	[1]	set1.4	real	SOAPfuse	51			56	0,91				
 40	[1]	set1.4	real	STAR-Fusion	49			56	0,88				
	[1]	set1.4	real	defuse	48			56	0,86				
	[1]	set1.5	real	Arriba	9			9	1,00				
	[1]	set1.5	real	FusionCatcher	3			9	0,33				
	[1]	set1.5	real	InFusion	5			9	0,56				
	[1]	set1.5	real	PRADA	0			9	0,00				
	[1]	set1.5	real	SOAPfuse	0			9	0,00				
	[1]	set1.5	real	STAR-Fusion	4			9	0,44				
	[1]	set1.5	real	defuse	2			9	0,22				
	[2]	set2.1	real	STAR-Fusion					0,69			0,42	
	[2]	set2.1	real	SeekFusion					0,89			0,58	
	[2]	set2.1	real	JAFFA-Hybrid					0,65			0,83	
	[2]	set2.1	real	Tophat-Fusion					0,01			0,83	
 10	[3]	set3.1	simulated	Arriba	67	39	106	80	0,84	0,63	0,72		
	[3]	set3.1	simulated	FuSeq				80					
	[3]	set3.1	simulated	InFusion	55	6	61	80	0,69	0,9	0,78		
	[3]	set3.1	simulated	pizzly	44	6	49	80	0,54	0,88	0,67		
	[3]	set3.1	simulated	STAR-Fusion	64	10	74	80	0,79	0,86	0,83		
	[3]	set3.1	simulated	STARChip	21	60	82	80	0,27	0,26	0,26		
	[3]	set3.1	simulated	EricScript	49	8	57	80	0,62	0,86	0,72		
	[3]	set3.1	simulated	FusionCatcher	52	16	68	80	0,65	0,76	0,70		
	[3]	set3.1	simulated	MapSplice	31	3	34	80	0,38	0,9	0,54		
	[3]	set3.1	simulated	ChimeraScan	71	11	82	80	0,88	0,86	0,87		
	[3]	set3.1	simulated	JAFFA	45	9	53	80	0,56	0,84	0,67		
	[3]	set3.1	simulated	ChimPipe	28	5	33	80	0,35	0,86	0,50		
	[3]	set3.1	simulated	SOAPfuse	59	10	69	80	0,74	0,85	0,79		
 10	[3]	set3.1	simulated	TopHat-Fusion	32	19	50	80	0,40	0,63	0,49		
	[3]	set3.1	simulated	INTEGRATE	61	11	72	80	0,77	0,85	0,81		
	[3]	set3.1	simulated	ChimerScope	22	32	54	80	0,28	0,41	0,33		
	[3]	set3.2	simulated	pizzly	196	72	268	250	0,78	0,73	0,76	0,1	5
	[3]	set3.2	simulated	FuSeq	211	124	335	250	0,84	0,63	0,72	0,5	8
	[3]	set3.2	simulated	STARChip	1	2	3	250	0,00	0,33	0,01	0,6	28
	[3]	set3.2	simulated	Arriba	233	66	299	250	0,93	0,78	0,85	1,0	30
	[3]	set3.2	simulated	STAR-Fusion	172	67	239	250	0,69	0,72	0,70	1,3	36
	[3]	set3.2	simulated	EricScript	158	258	416	250	0,63	0,38	0,47	4,7	5
	[3]	set3.2	simulated	SOAPfuse	204	48	252	250	0,82	0,81	0,81	5,3	6
	[3]	set3.2	simulated	INTEGRATE	143	34	177	250	0,57	0,81	0,67	5,5	25
	[3]	set3.2	simulated	InFusion	166	42	208	250	0,66	0,8	0,73	7,0	5
	[3]	set3.2	simulated	JAFFA	133	360	493	250	0,53	0,27	0,36	7,7	40
 10	[3]	set3.2	simulated	ChimerScope	56	744	800	250	0,22	0,07	0,11	8,5	56
	[3]	set3.2	simulated	MapSplice	158	2	160	250	0,63	0,99	0,77	12,2	6
	[3]	set3.2	simulated	TopHat-Fusion	102	83	185	250	0,41	0,55	0,47	24,3	16
	[3]	set3.2	simulated	ChimPipe	190	14	204	250	0,76	0,93	0,84	24,7	8
	[3]	set3.2	simulated	FusionCatcher	150	42	192	250	0,60	0,78	0,68	27,0	15
	[3]	set3.2	simulated	ChimeraScan	204	319	523	250	0,82	0,39	0,53	28,6	4
	[3]	set3.3	simulated	Arriba	227	68	295	250	0,91	0,77	0,83		
	[3]	set3.3	simulated	FuSeq	208	89	297	250	0,83	0,7	0,76		
	[3]	set3.3	simulated	InFusion	162	41	203	250	0,65	0,8	0,72		
	[3]	set3.3	simulated	pizzly	198	70	268	250	0,79	0,74	0,77		
	[3]	set3.3	simulated	STAR-Fusion	175	79	254	250	0,70	0,69	0,69		
	[3]	set3.3	simulated	STARChip	43	76	119	250	0,17	0,36	0,23		
	[3]	set3.3	simulated	EricScript	160	433	593	250	0,64	0,27	0,38		
 10	[3]	set3.3	simulated	FusionCatcher	161	43	204	250	0,64	0,79	0,71		
	[3]	set3.3	simulated	MapSplice	162	2	164	250	0,65	0,99	0,78		
	[3]	set3.3	simulated	ChimeraScan	205	295	500	250	0,82	0,41	0,55		
	[3]	set3.3	simulated	JAFFA	177	455	632	250	0,71	0,28	0,40		
	[3]	set3.3	simulated	ChimPipe	188	14	202	250	0,75	0,93	0,83		
	[3]	set3.3	simulated	SOAPfuse	202	47	249	250	0,81	0,81	0,81		
	[3]	set3.3	simulated	TopHat-Fusion	89	70	159	250	0,36	0,56	0,44		
	[3]	set3.3	simulated	INTEGRATE	147	46	193	250	0,59	0,76	0,66		
	[3]	set3.3	simulated	ChimerScope	86	1143	1229	250	0,34	0,07	0,12		
	[3]	set3.4	simulated	Arriba	226	60	286	250	0,90	0,79	0,84		
	[3]	set3.4	simulated	FuSeq	207	93	300	250	0,83	0,69	0,75		
	[3]	set3.4	simulated	InFusion	169	42	211	250	0,68	0,8	0,73		
	[3]	set3.4	simulated	pizzly	197	59	256	250	0,79	0,77	0,78		
 24	[3]	set3.4	simulated	STAR-Fusion	169	83	252	250	0,68	0,67	0,67		
	[3]	set3.4	simulated	STARChip	43	100	143	250	0,17	0,3	0,22		
	[3]	set3.4	simulated	EricScript	150	427	577	250	0,60	0,26	0,36		
	[3]	set3.4	simulated	FusionCatcher	165	64	229	250	0,66	0,72	0,69		
	[3]	set3.4	simulated	MapSplice	152	2	154	250	0,61	0,99	0,75		
	[3]	set3.4	simulated	ChimeraScan	200	300	500	250	0,80	0,4	0,53		
	[3]	set3.4	simulated	JAFFA	182	468	650	250	0,73	0,28	0,40		
	[3]	set3.4	simulated	ChimPipe	177	15	192	250	0,71	0,92	0,80		
	[3]	set3.4	simulated	SOAPfuse	199	44	243	250	0,80	0,82	0,81		
	[3]	set3.4	simulated	TopHat-Fusion	68	51	119	250	0,27	0,57	0,37		
	[3]	set3.4	simulated	INTEGRATE	139	37	176	250	0,56	0,79	0,65		
	[3]	set3.4	simulated	ChimerScope	85	978	1063	250	0,34	0,08	0,13		
	[3]	set3.5	real	pizzly	20	981	1001	62	0,32	0,02	0,04	0,2	5
[3]	set3.												

[3]	set3.6	real	STARChip	6	80	86	27	0,22	0,07	0,11
[3]	set3.6	real	EricScript	16	107	123	27	0,59	0,13	0,21
[3]	set3.6	real	FusionCatcher	20	24	44	27	0,74	0,45	0,56
[3]	set3.6	real	MapSplice	17	22	39	27	0,63	0,44	0,52
[3]	set3.6	real	ChimeraScan	23	2277	2300	27	0,85	0,01	0,02
[3]	set3.6	real	JAFFA	17	68	85	27	0,63	0,2	0,30
[3]	set3.6	real	ChimPipe	18	42	60	27	0,67	0,3	0,41
[3]	set3.6	real	SOAPfuse	20	98	118	27	0,74	0,17	0,28
[3]	set3.6	real	TopHat-Fusion	12	28	40	27	0,44	0,3	0,36
[3]	set3.6	real	INTEGRATE	16	34	50	27	0,59	0,32	0,42
[3]	set3.6	real	ChimerAScope	10	133	143	27	0,37	0,07	0,12
[3]	set3.7	real	Arriba	45	110	155	99	0,46	0,29	0,35
[3]	set3.7	real	FuSeq	35	50	85	99	0,35	0,41	0,38
[3]	set3.7	real	InFusion	25	38	63	99	0,25	0,4	0,31
[3]	set3.7	real	pizzly	41	984	1025	99	0,41	0,04	0,07
[3]	set3.7	real	STAR-Fusion	41	34	75	99	0,41	0,55	0,47
[3]	set3.7	real	STARChip	8	81	89	99	0,08	0,09	0,09
[3]	set3.7	real	EricScript	23	98	121	99	0,23	0,19	0,21
[3]	set3.7	real	FusionCatcher	27	17	44	99	0,27	0,61	0,38
[3]	set3.7	real	MapSplice	21	18	39	99	0,21	0,54	0,30
[3]	set3.7	real	ChimeraScan	41	1325	1366	99	0,41	0,03	0,06
[3]	set3.7	real	JAFFA	22	63	85	99	0,22	0,26	0,24
[3]	set3.7	real	ChimPipe	28	32	60	99	0,28	0,47	0,35
[3]	set3.7	real	SOAPfuse	36	84	120	99	0,36	0,3	0,33
[3]	set3.7	real	TopHat-Fusion	15	25	40	99	0,15	0,38	0,22
[3]	set3.7	real	INTEGRATE	26	24	50	99	0,26	0,52	0,35
[3]	set3.7	real	ChimerAScope	17	125	142	99	0,17	0,12	0,14
[4]	set4.1	simulated	MetaFusion.top_3	1989	90	2079	2500	0,796	0,96	0,87
[4]	set4.1	simulated	MetaFusion	2091	164	2255	2500	0,836	0,93	0,88
[4]	set4.1	simulated	Arriba	2104	273	2377	2500	0,842	0,89	0,86
[4]	set4.1	simulated	STAR-Fusion	1929	117	2046	2500	0,772	0,94	0,85
[4]	set4.1	simulated	STAR-SEQR	1953	306	2259	2500	0,781	0,87	0,82
[4]	set4.1	simulated	defuse	1910	179	2089	2500	0,764	0,91	0,83
[4]	set4.1	simulated	EricScript	1408	68	1476	2500	0,563	0,95	0,71
[4]	set4.1	simulated	INTEGRATE	1727	325	2052	2500	0,691	0,84	0,76
[4]	set4.1	simulated	FusionMap	1500	122	1622	2500	0,6	0,93	0,73
[4]	set4.2	simulated	MetaFusion.top_3	2333	86	2419	2500	0,933	0,96	0,95
[4]	set4.2	simulated	MetaFusion	2389	147	2536	2500	0,956	0,94	0,95
[4]	set4.2	simulated	Arriba	2304	260	2564	2500	0,922	0,90	0,91
[4]	set4.2	simulated	STAR-Fusion	2250	170	2420	2500	0,9	0,93	0,92
[4]	set4.2	simulated	STAR-SEQR	2337	306	2643	2500	0,935	0,88	0,91
[4]	set4.2	simulated	defuse	2191	312	2503	2500	0,876	0,88	0,88
[4]	set4.2	simulated	EricScript	1722	445	2167	2500	0,689	0,80	0,74
[4]	set4.2	simulated	INTEGRATE	1708	369	2077	2500	0,683	0,82	0,75
[4]	set4.2	simulated	FusionMap	1986	81	2067	2500	0,794	0,96	0,87
[4]	set4.3	simulated	MetaFusion.top_3	56	3	59	61	0,918	0,95	0,93
[4]	set4.3	simulated	MetaFusion	57	12	69	61	0,934	0,83	0,88
[4]	set4.3	simulated	Arriba	56	17	73	61	0,918	0,77	0,84
[4]	set4.3	simulated	STAR-Fusion	50	11	61	61	0,82	0,82	0,82
[4]	set4.3	simulated	STAR-SEQR	57	33	90	61	0,934	0,63	0,76
[4]	set4.3	simulated	defuse	51	231	282	61	0,836	0,18	0,30
[4]	set4.3	simulated	EricScript	40	262	302	61	0,656	0,13	0,22
[4]	set4.3	simulated	INTEGRATE	45	186	231	61	0,738	0,20	0,31
[4]	set4.3	simulated	FusionMap	42	12	54	61	0,689	0,78	0,73
[4]	set4.4	real	MetaFusion.top_3	40	27	67	53	0,755	0,60	0,67
[4]	set4.4	real	MetaFusion	42	37	79	53	0,792	0,53	0,64
[4]	set4.4	real	Arriba	42	148	190	53	0,792	0,22	0,35
[4]	set4.4	real	STAR-Fusion	41	31	72	53	0,774	0,57	0,66
[4]	set4.4	real	STAR-SEQR	39	119	158	53	0,736	0,25	0,37
[4]	set4.4	real	defuse	34	123	157	53	0,642	0,22	0,32
[4]	set4.4	real	EricScript	29	118	147	53	0,547	0,20	0,29
[4]	set4.4	real	INTEGRATE	37	39	76	53	0,698	0,49	0,57
[4]	set4.4	real	FusionMap	30	74	104	53	0,566	0,29	0,38
[4]	set4.5	real	MetaFusion.top_3	16	10	26	17	0,941	0,62	0,74
[4]	set4.5	real	MetaFusion	17	26	43	17	1	0,40	0,57
[4]	set4.5	real	Arriba	15	314	329	17	0,882	0,05	0,09
[4]	set4.5	real	STAR-Fusion	10	37	47	17	0,588	0,21	0,31
[4]	set4.5	real	STAR-SEQR	14	90	104	17	0,824	0,14	0,23
[4]	set4.5	real	defuse	16	327	343	17	0,941	0,05	0,09
[4]	set4.5	real	EricScript	9	343	352	17	0,529	0,03	0,05
[4]	set4.5	real	INTEGRATE	11	39	50	17	0,647	0,22	0,33
[4]	set4.5	real	FusionMap	11	29	40	17	0,647	0,28	0,39
[4]	set4.6	real	MetaFusion.top_3	12	1	13	12	1	0,92	0,96
[4]	set4.6	real	MetaFusion	12	1	13	12	1	0,92	0,96
[4]	set4.6	real	Arriba	12	8	20	12	1	0,60	0,75
[4]	set4.6	real	STAR-Fusion	11	3	14	12	0,917	0,79	0,85
[4]	set4.6	real	STAR-SEQR	12	3	15	12	1	0,80	0,89
[4]	set4.6	real	defuse	10	6	16	12	0,833	0,63	0,71
[4]	set4.6	real	EricScript	10	23	33	12	0,833	0,30	0,44
[4]	set4.6	real	INTEGRATE	5	5	10	12	0,417	0,50	0,46
[4]	set4.6	real	FusionMap	12	3	15	12	1	0,80	0,89
[5]	set5.1	simulated	STAR-SEQR	445	2		500	0,89	1	0,94
[5]	set5.1	simulated	ARRIBA	443	11		500	0,89	0,98	0,93
[5]	set5.1	simulated	STAR-Fusion	422	9		500	0,84	0,98	0,90
[5]	set5.1	simulated	JAFFA-Direct	421	13		500	0,84	0,97	0,90
[5]	set5.1	simulated	nfuse	415	15		500	0,83	0,97	0,89
[5]	set5.1	simulated	PIZZLY	420	21		500	0,84	0,95	0,89
[5]	set5.1	simulated	TopHat-Fusion	403	5		500	0,81	0,99	0,89
[5]	set5.1	simulated	deFuse	394	8		500	0,79	0,96	0,87
[5]	set5.1	simulated	ARRIBA_hc	389	2		500	0,78	0,99	0,87
[5]	set5.1	simulated	FUSIONCATCHER	387	8		500	0,77	0,98	0,86
[5]	set5.1	simulated	InFusion	378	4		500	0,76	0,99	0,86
[5]	set5.1	simulated	MapSplice	363	1		500	0,73	1	0,84
[5]	set5.1	simulated	STARCHIP	363	3		500	0,73	0,99	0,84
[5]	set5.1	simulated	TrinityFusion-C	358	2		500	0,72	0,99	0,83
[5]	set5.1	simulated	ChimeraScan	416	84		500	0,83	0,83	0,83
[5]	set5.1	simulated	JAFFA-Hybrid	360	10		500	0,72	0,97	0,83
[5]	set5.1	simulated	SOAP-fuse	344	2		500	0,69	0,99	0,81
[5]	set5.1	simulated	TrinityFusion-UC	332	1		500	0,66	1	0,80
[5]	set5.1	simulated	EricScript	343	31		500	0,69	0,92	0,79
[5]	set5.1	simulated	JAFFA-Assembly	333	68		500	0,67	0,83	0,74
[5]	set5.1	simulated	PRADA	278	1		500	0,56	1	0,72
[5]	set5.1	simulated	ChimPipe	281	2		500	0,56	0,99	0,72
[5]	set5.1	simulated	TrinityFusion-D	200	2		500	0,4	0,99	0,57
[5]	set5.1	simulated	FusionHunter	183	5		500	0,37	0,97	0,54
[5]	set5.2	simulated	STAR-SEQR	2339	10		2500	0,94	1	0,97
[5]	set5.2	simulated	ARRIBA	2335	53		2500	0,93	0,98	0,95
[5]	set5.2	simulated	STAR-Fusion	2293	45		2500	0,92	0,98	0,95
[5]	set5.2	simulated	PIZZLY	2266	129		2500	0,91	0,95	0,93
[5]	set5.2	simulated	TopHat-Fusion	2116	30		2500	0,85	0,99	0,91
[5]	set5.2	simulated	nfuse	2170	94		2500	0,87	0,96	0,91
[5]	set5.2	simulated	JAFFA-Direct	2146	56		2500	0,86	0,97	0,91
[5]	set5.2	simulated	InFusion	2068	17		2500	0,83	0,99	0,90
[5]	set5.2	simulated	FUSIONCATCHER	2075	38		2500	0,83	0,98	0,90
[5]	set5.2	simulated	ARRIBA_hc	2034	15		2500	0,81	0,99	0,89
[5]	set5.2	simulated	deFuse	2064	78		2500	0,83	0,96	0,89
[5]	set5.2	simulated	STARCHIP	2004	10		2500	0,8	1	0,89
[5]	set5.2	simulated	MapSplice	1967	6		2500	0,79	1	0,88
[5]	set5.2	simulated	TrinityFusion-C	1963	12		2500	0,79	0,99	0,88
[5]	set5.2	simulated	ChimPipe	1953	5		2500	0,78	1	0,88
[5]	set5.2	simulated	TrinityFusion-UC	1892	11		2500	0,76	0,99	0,86
[5]	set5.2	simulated	ChimeraScan	2094	342		2500	0,84	0,86	0,85
[5]	set5.2	simulated	JAFFA-Hybrid	1831	54		2500	0,73	0,97	0,83
[5]	set5.2	simulated	SOAP-fuse	1769	21		2500	0,71	0,99	0,83
[5]	set5.2	simulated	EricScript	1791	201		2500	0,72	0,9	0,80

[5]	set5.2	simulated	PRADA	1643	2	2500	0,66	1	0,80	
[5]	set5.2	simulated	JAFFA-Assembly	1620	228	2500	0,65	0,88	0,75	
[5]	set5.2	simulated	TrinityFusion-D	1097	12	2500	0,44	0,99	0,61	
[5]	set5.2	simulated	FusionHunter	1091	17	2500	0,44	0,98	0,61	
[5]	set5.3	real	STAR-Fusion							1
[5]	set5.3	real	ARRIBA_hc	341	106	447	0,70	0,76	0,73	
[5]	set5.3	real	Arriba	394	241	635	0,80	0,62	0,70	2
[5]	set5.3	real	Pizzly	235	303	538	0,48	0,44	0,46	2
[5]	set5.3	real	STARChip	229	81	310	0,47	0,74	0,57	2
[5]	set5.3	real	STAR-SEQR	429	301	730	0,88	0,59	0,70	3
[5]	set5.3	real	STAR-Fusion	411	109	520	0,84	0,79	0,81	4
[5]	set5.3	real	FusionCatcher			0	0,00			4
[5]	set5.3	real	FusionCatcher	325	327	652	0,66	0,50	0,57	4
[5]	set5.3	real	EricScript	157	1471	1628	0,32	0,10	0,15	10
[5]	set5.3	real	nFuse	226	1328	1554	0,46	0,15	0,22	10
[5]	set5.3	real	FusionHunter	97	58	155	0,20	0,63	0,30	11
[5]	set5.3	real	PRADA	188	21	209	0,38	0,90	0,54	15
[5]	set5.3	real	JEFFA-Direct	375	711	1086	0,77	0,35	0,48	15
[5]	set5.3	real	deFuse	196	911	1107	0,40	0,18	0,25	20
[5]	set5.3	real	ChimPipe	270	8	278	0,55	0,97	0,70	25
[5]	set5.3	real	SOAP-fuse	246	56	302	0,50	0,81	0,62	25
[5]	set5.3	real	TopHat-Fusion	319	390	709	0,65	0,45	0,53	25
[5]	set5.3	real	ChimeraScan	145	435	580	0,30	0,25	0,27	30
[5]	set5.3	real	MapSplice	315	283	598	0,64	0,53	0,58	30
[5]	set5.3	real	JEFFA-Assembly	273	296	569	0,56	0,48	0,52	30
[5]	set5.3	real	TrinityFusion-C	274	16	290	0,56	0,94	0,70	45
[5]	set5.3	real	InFusion	318	334	652	0,65	0,49	0,56	60
[5]	set5.3	real	JEFFA-Hybrid	291	1185	1476	0,59	0,20	0,30	65
[5]	set5.3	real	TrinityFusion-D	216	24	240	0,44	0,90	0,59	70
[5]	set5.3	real	TrinityFusion-UC	262	34	296	0,53	0,89	0,67	70
[6]	set6.1	real	FuSeq	22	31	53	0,81	0,42	0,55	1,83
[6]	set6.1	real	FusionMap	12	35	47	0,44	0,26	0,32	4,08
[6]	set6.1	real	TRUP	0	0	0	0,00			136,04
[6]	set6.1	real	TopHat-Fusion	24	237	261	0,89	0,09	0,17	
[6]	set6.1	real	SOAPfuse	24	37	61	0,89	0,39	0,55	
[6]	set6.1	real	JAFFA	20	22	42	0,74	0,48	0,58	
[6]	set6.2	real	FuSeq	22	31	53	0,99	0,22	0,29	
[6]	set6.2	real	FusionMap	22	25	47	0,99	0,22	0,30	
[6]	set6.2	real	TRUP	0	0	0	0,00			
[6]	set6.2	real	TopHat-Fusion	35	226	261	0,99	0,35	0,19	
[6]	set6.2	real	SOAPfuse	41	20	61	0,99	0,41	0,51	
[6]	set6.2	real	JAFFA	28	14	42	0,99	0,28	0,40	
[6]	set6.3	real	FuSeq	7	14	21	0,11	0,64	0,33	0,44
[6]	set6.3	real	FusionMap	3	16	19	0,11	0,27	0,16	0,20
[6]	set6.3	real	TRUP	0	0	0	0,00			
[6]	set6.3	real	TopHat-Fusion	4	25	29	0,11	0,36	0,14	0,20
[6]	set6.3	real	SOAPfuse	10	98	108	0,11	0,91	0,09	0,17
[6]	set6.3	real	JAFFA	2	2	4	0,11	0,18	0,50	0,27
[6]	set6.4	real	FuSeq	29	159	188	0,31	0,94	0,15	0,26
[6]	set6.4	real	FusionMap	29	162	191	0,31	0,94	0,15	0,26
[6]	set6.4	real	TRUP	20	189	209	0,31	0,65	0,10	0,17
[6]	set6.4	real	TopHat-Fusion	29	279	308	0,31	0,94	0,09	0,17
[6]	set6.4	real	SOAPfuse	22	277	289	0,31	0,71	0,07	0,13
[6]	set6.4	real	JAFFA	30	874	904	0,31	0,97	0,03	0,06
[7]	set7.1	real	Gfusion	28	6	34	0,37	0,76	0,82	0,79
[7]	set7.1	real	TopHat-Fusion	30	36	66	0,37	0,81	0,45	0,58
[7]	set7.1	real	FusionMap	16	10	26	0,37	0,43	0,62	0,51
[7]	set7.1	real	JAFFA	21	16	37	0,37	0,57	0,57	0,57
[7]	set7.1	real	ChimPipe	30	8	38	0,37	0,81	0,79	0,80
[7]	set7.2	realSE	Gfusion	6	28	34			0,18	0,00
[7]	set7.2	realSE	TopHat-Fusion	3	71	74			0,04	0,00
[7]	set7.2	realSE	FusionMap	5	221	226			0,02	0,00
[7]	set7.2	realSE	JAFFA							
[7]	set7.2	realSE	ChimPipe							
[7]	set7.3	simulated	Gfusion	37	7	44	0,42	0,88	0,84	0,86
[7]	set7.3	simulated	TopHat-Fusion	35	50	85	0,42	0,83	0,41	0,55
[7]	set7.3	simulated	FusionMap	22	160	182	0,42	0,52	0,12	0,20
[7]	set7.3	simulated	JAFFA	22	77	99	0,42	0,52	0,22	0,31
[7]	set7.3	simulated	ChimPipe	9	13	22	0,42	0,21	0,41	0,28
[8]	set8.1	real	INTEGRATE C				0,138	0,89	0,81	0,85
[8]	set8.1	real	INTEGRATE G				0,138	0,67	0,85	0,75
[8]	set8.1	real	INTEGRATE T				0,138	0,46	0,93	0,62
[8]	set8.1	real	INTEGRATE S				0,138	0,42	0,85	0,56
[8]	set8.1	real	nFuse	45			0,138	0,33	0,44	0,38
[8]	set8.1	real	ChimeraScan				0,138	0,29	0,74	0,42
[8]	set8.1	real	FusionCatcher				0,138	0,25	0,95	0,40
[8]	set8.1	real	TopHat-Fusion				0,138	0,1	1,00	0,18
[8]	set8.1	real	pyPRADA				0,138	0,1	1,00	0,18
[8]	set8.1	real	TRUP				0,138	0,1	1,00	0,18
[8]	set8.1	real	Comrad				0,138	0,1	1,00	0,18
[8]	set8.1	real	BreakTrans				0,138	0,1	1,00	0,18
[9]	set9.1	real	FusionMap	0			0,44	0,00		0,72
[9]	set9.1	real	EricScript	26			0,44	0,59		4,20
[9]	set9.1	real	FusionCatcher	3			0,44	0,07		6,05
[9]	set9.1	real	Chimerascan	31			0,44	0,70		9,50
[9]	set9.1	real	FusionHunter	9			0,44	0,20		12,12
[9]	set9.1	real	nFuse	0			0,44	0,00		12,70
[9]	set9.1	real	SOAPfuse	0			0,44	0,00		15,68
[9]	set9.1	real	TopHat-Fusion	0			0,44	0,00		32,57
[9]	set9.1	real	BreakFusion	5			0,44	0,11		38,97
[9]	set9.1	real	MapSplice	0			0,44	0,00		43,73
[9]	set9.1	real	JAFFA	1			0,44	0,02		66,27
[9]	set9.1	real	Bellerophontes	0			0,44	0,00		
[9]	set9.2	simulated	Chimerascan	4			0,50	0,08		0,05
[9]	set9.2	simulated	EricScript	39			0,50	0,78		0,05
[9]	set9.2	simulated	MapSplice	43			0,50	0,86		0,05
[9]	set9.2	simulated	Bellerophontes	33	42		0,50	0,66	0,79	0,10
[9]	set9.2	simulated	JAFFA	44			0,50	0,88		0,12
[9]	set9.2	simulated	FusionHunter	18			0,50	0,36		0,15
[9]	set9.2	simulated	FusionCatcher	33			0,50	0,66		0,18
[9]	set9.2	simulated	TopHat-Fusion	26			0,50	0,52		0,25
[9]	set9.2	simulated	FusionMap	28			0,50	0,56		0,32
[9]	set9.2	simulated	nFuse	15			0,50	0,30		0,77
[9]	set9.2	simulated	SOAPfuse	41			0,50	0,82		0,87
[9]	set9.2	simulated	BreakFusion	2			0,50	0,04		
[10]	set10.1	simulated	JAFFA - Hybrid	44	0	44	0,50	0,88	1,00	0,94
[10]	set10.1	simulated	JAFFA - Assembly	39	0	39	0,50	0,78	1,00	0,88
[10]	set10.1	simulated	SOAPfuse	37	1	38	0,50	0,74	0,97	0,84
[10]	set10.1	simulated	JAFFA - Direct	34	0	34	0,50	0,68	1,00	0,81
[10]	set10.1	simulated	deFuse	34	0	34	0,50	0,68	1,00	0,81
[10]	set10.1	simulated	TopHat-Fusion	27	0	27	0,50	0,54	1,00	0,70
[10]	set10.1	simulated	FusionCatcher	0	0	0	0,50			

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benchmark_number	year	reference	number_of_tools	tool_name	tool_version	tool_notes	analysis_parameters	time	memory_consumption	number_of_datasets	git_code	ppv_tpr_source	notes
1	2019	[5]	23	STAR-Fusion	v1.5 kickstart								
				ARRIBA_hc									
				Arriba	1.1.0.								
				Pizzly	0.37.3								
				STARChip	May012019								
				STAR-SEQR	v0.6.7.								
				STAR-Fusion	v1.5								
				FusionCatcher	V0994e								
				FusionCatcher	v1.10								
				EricScript	0.5.3								
				nFuse	0.2.1								
				FusionHunter	v1.4								
				PRADA	v1.2								
				JAFFA-Direct	v1.06	doesn't work on PE50, leverage	used its own recommended alignment and analysis strategy and parameters, as implemented in its respective package	time ranking	no info	3	Docker, easy to reuse	Docker	
				deFuse	0.6.1								
				ChimPipe	0.9.5								
2	2021	[1]	7	SOAP-fuse	v1.26								
				TopHat-Fusion	2.0.10								
				ChimeraScan	0.4.5								
				MapSplice	v2.2.0								
				JAFFA-Assembly	v1.06								
				TrinityFusion-C	v0.2.0	only chimeric reads							
				InFusion									
				JAFFA-Hybrid	v1.06	doesn't work on PE50							
				TrinityFusion-D	v0.2.0	involving assembly of all input							
				TrinityFusion-UC	v0.2.0	both unmapped and chimeric							
				Arriba	v1.00								
				FusionCatcher	v1.00								
				InFusion	v0.8		tools were run with default parameters	time ranking	memory ranking	4	GitHub	Excel, need to calculate	
				PRADA	v1.2								
				SOAPfuse	v1.27								
				STAR-Fusion	v1.40								
				deFuse	v0.8.1								
3	2016	[8]	12	Bellerophon	version 0.4.0								
				BreakFusion									
				Chimerascan	version 0.4.5								
				nFuse	version 0.2.1								
				EricScript	version 0.5.1								
				FusionCatcher	v0.99.4c	doesn't work on PE50	run using a default configuration of each tool with SLURM (Simple Linux Utility for Resource Management) scripts	time ranking	memory ranking	4	no info	html, Excel	
				FusionHunter	v1.4-Linux_x86_64	doesn't work on PE50							
				FusionMap	FusionMap_2015-03-1	doesn't work on PE50							
				JAFFA	version-1.06	doesn't work on PE50							
				MapSplice	v2.1.9								
				SOAPfuse	v1.26								
				TopHat-Fusion	2.1.0.Linux_x86_64	doesn't work on PE50							
				SeekFusion									
				STAR-Fusion									
				JAFFA-Hybrid	version-1.09		default parameters	time ranking	no info	1	Docker	info in text	
				TopHat-Fusion	2.1.0.								
4	2017	[7]	5	Gfusion		works on SE							
				TopHat-Fusion		works on SE							
				FusionMap		works on SE	default parameters			3			
				JAFFA									
				ChimPipe									
				FuSeq									
				FusionMap									
				TRUP									
				TopHat-Fusion			default parameters	for one set	for one set	4			
				SOAPfuse									
				JAFFA									
				Arriba									
				FuSeq									
				InFusion									
				pizzly									
				STAR-Fusion									
5	2021	[3]	16	STARChip									
				EricScript									
				FusionCatcher		supplementary file 1	used default parameters (or default configuration file)	for some sets	for some sets	4			
				MapSplice									
				ChimeraScan									
				JAFFA									
				ChimPipe									
				SOAPfuse									
				TopHat-Fusion									
				INTEGRATE									
				ChimeRScope									
				deFuse									
				Arriba									
				STAR-SEQR									
				STAR-Fusion			default parameters	no info	no info	7	based on Haas's Benchmarking tool	EXCEL	
				INTEGRATE									
6	2021	[4]	8	EricScript									
				FusionMap									
				MetaFusion									
				INTEGRATE C		WGS and RNA-seq caller							
				INTEGRATE G									
				INTEGRATE T									
				INTEGRATE S									
				nFuse		WGS and RNA-seq caller							
				ChimeraScan									
				FusionCatcher			default parameters	no info	no info	1			
				TopHat-Fusion									
				pyPRADA									
				TRUP									
				Comrad		WGS and RNA-seq caller							
				BreakTrans		WGS and RNA-seq caller							
7	2015	[10]	5	JAFFA		works on SE							
				TopHat-Fusion 2.0.13	2.0.13	works on SE							
				DeFuse 0.6.2	0.6.2		parameters in additional file	no info	no info	7			
				SOAPfuse 1.26	v1.26								
				FusionCatcher 0.99.3d	0.99.3d	the best for 50bp							

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id	mark	reference	year	number_of_datasets	real_simulated	transcriptome_panel	PE_SE	read_length	size_of_dataset	cancer_type	validation_gold_standard
1	1				simulated	not applicable	no info	75	40000000	not applicable	not applicable
7	7				spike-in	not applicable	PE	101	154000000	synthetic RNA molecules into RNA libraries of the melanoma cell line	WGS if available, experimental validation using Sanger sequencing
7	1	[1]	2021	4	real (cells)	no info	PE	101	297000000	including the breast cancer cell line MCF-7	WGS if available, experimental validation using Sanger sequencing
7	1				real (patients)	no info	no info	no info	no info	prostate cancer, diffuse large B-cell lymphoma	WGS if available, experimental validation using Sanger sequencing
2	2										The NGS assay results were confirmed by RT-PCR and FISH tests;
2	1	[2]	2021	1	real (patients)	panel 80 genes	PE	no info	no info	sarcoma and neurological cancer clinical cases	chromosomal microarray (CMA) and reverse transcription polymerase chain reaction (RT-PCR) with or without subsequent direct Sanger sequencing
42	3		2021	4	simulated	not applicable	PE	76	307920	not applicable	not applicable
35	3				simulated	not applicable	PE	50, 76, 101	135000000	not applicable	not applicable
35	3				real (cells)	no info	PE	101	120000000	human prostate cancer	previous research
35	3				real (cells)	no info	PE	50	87000000	breast cancer cell	previous research
27	4		2021	7	simulated	not applicable	PE	101	195000000	not applicable	not applicable
27	4				simulated	not applicable	no info	no info	no info	not applicable	not applicable
27	4				simulated	not applicable	PE	50, 100	60000000	not applicable	not applicable
27	4				real (cells)	transcriptome	PE	50, 100	20000000	breast cancer	previous research
27	4				real (patients)	transcriptome	no info	no info	no info	melanoma and CML	previous research
27	4				real (patients)	in el TruSight 1385 gen	no info	no info	no info	NTRK	previous research
27	4				real	real	PE	50, 100	no info	prostate cancer	previous research
1	1				simulated	not applicable	PE	50	30000000	not applicable	not applicable
7	1				simulated	not applicable	PE	101	30000000	not applicable	not applicable
7	5	[5]	2019	3	real (cells)	transcriptome	PE	100 or mixture with 50	20000000	cell lines: breast cancer, prostate cancer, lung adenocarcinoma, erythroleukemia	a "wisdom of crowds" approach
27	6		2018	4	real (cells)	no info	PE	50	9000000	breast-cancer	previous research
27	6				real (patients)	no info	PE	50	14000000	melanoma	previous research
27	6				real (patients)	no info	PE	101	25000000	glioma	previous research
27	6				simulated	not applicable	PE	100	112000000	not applicable	previous research
10	7		2017	3	real (cells)	no info	PE	no info	no info	breast cancer cell lines	previous research
10	7				real (cells)	no info	SE	76	14000000	chronic myeloid leukemia cells	previous research
10	7				simulated	no info	PE	76	24000000	not applicable	not applicable
10	7				simulated	not applicable	PE	75	57209	not applicable	not applicable
10	7				simulated	not applicable	PE	50, 75, 100	70000000	not applicable	not applicable
10	7				simulated	not applicable	PE	75	70057209	not applicable	not applicable
10	7	[8]	2016	4	real (cells)	no info	PE	50, 100	62117396	prostate cell line	confirmed by Sanger's sequencing
1	9		2016	1	real (cells)	transcriptome	PE	no info	1884000000	breast cancer	WGS (PE, 321000000 reads), qRT-PCR
34	10				simulated	not applicable	PE	75	57000	not applicable	not applicable
34	10				simulated	not applicable	PE	250	5000000	not applicable	not applicable
34	10				real (cells)	no info	SE	50, 100	30000000, 40000000	breast cancer	no info
34	10				real (cells)	no info	PE	50, 100	10000000, 20000000	breast cancer	no info
34	10				simulated	not applicable	PE	100	20000000	not applicable	previous research
34	10				real (cells)	no info	PE	50	38000000	breast cancer	previous research
34	10				real (cells)	no info	PE	100	40000000	breast cancer	previous research
34	10				real (cells)	no info	PE	no info	44531	breast cancer	previous research
34	10				real	no info	PE	no info	no info	glioma	previous research

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Benchmark number	1	2	3	4	5	6	7	8	9	10	
Reference	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	
Tool and reference											sum
Arriba [1]	1		1	1	1						4
Bellerophon [11]									1		1
BreakFusion [12]									1		1
BreakTrans [13]								1			1
ChimeraScan [14]			1		1			1	1		4
ChimeRScope [15]			1								1
ChimPipe [16]			1		1		1				4
Comrad [17]								1			1
deFuse [18]	1			1	1						3
EricScript [19]			1	1	1				1		4
FuSeq [20]			1			1					2
FusionCatcher [21]	1		1		1			1	1		5
FusionHunter [22]					1				1		2
FusionMap [23]				1		1	1		1		5
Gfusion [7]							1				2
InFusion [24]	1		1		1						3
INTEGRATE [8]			1	1							2
INTEGRATE C [8]								1			1
INTEGRATE G [8]								1			1
INTEGRATE S [8]								1			1
INTEGRATE T [8]								1			1
JAFFA [10]			1			1	1		1		5
JAFFA-Assembly [10]					1						1
JAFFA-Direct [10]					1						1
JAFFA-Hybrid [10]		1			1						2
MapSplice [25]			1		1				1		3
MetaFusion [4]				1					1		1
nFuse [26]					1			1			3
Pizzly [27]			1		1				1		2
PRADA [28]	1				1						2
pyPRADA [28]								1			1
SeekFusion [2]		1									1
SOAPfuse [29]	1		1		1	1			1		5
STAR-Fusion [5]	1	1	1	1	1						5
STAR-SEQR [30]				1	1						2
STARChip [31]			1		1						2
TopHat-Fusion [32]		1	1		1	1	1	1	1		8
TrinityFusion-C [33]					1						1
TrinityFusion-C [33]					1						1
TrinityFusion-UC [33]					1						1
TRUP [34]						1		1			2
sum	7	4	16	8	23	6	5	12	12	5	

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